

Eshan Raul

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Education

University of Wisconsin–Madison

August 2024–May 2029

Ph.D. in Astronomy

University of Michigan, Ann Arbor

August 2020–May 2024

B.S. in Honors Astronomy/Astrophysics

Research Interests

Astrobiology & habitability

Exoplanetary atmospheres/dynamics

Astrochemistry

Experience

Research at University of Wisconsin–Madison

January 2026–

Working with Professor Thomas Beatty on investigating the trends of cloud presence in the super-Earth and sub-Neptune exoplanet population through JWST archival data using machine learning

Research at University of Wisconsin–Madison

September 2025–

Working with Professor Juliette Becker on identifying the spin-orbit misalignment of several super-Earth and sub-Neptune exoplanetary systems through measurements of the Rossiter-McLaughlin effect

JWST General Observing Program 4098

January 2024–

Working with Dr. Ryan MacDonald on the identification of five potential water worlds by determining their atmospheric compositions via transmission spectra analysis on the team of a Large JWST Cycle 2 Program

JWST General Observing Program 3034

September 2024–March 2026

Working with Professor Ke (Coco) Zhang to constrain the abundances of complex organic molecules in the terrestrial-planet-forming region of protoplanetary disks on the team of a Large JWST Cycle 2 Program

Research at University of Michigan

January 2022–January 2024

Worked with Professor Edwin Bergin on the chemical evolution of organic molecules in the terrestrial planet-forming regions of protoplanetary disks through modeling and data interpretation of carbon grain destruction

Research Capstone

September–December 2023

Worked with Professor Emily Rauscher on investigation of the influence of exoplanet rotation on hot Jupiter emission spectra to determine if this effect is relevant to JWST-level precision measurements

Research at University of Michigan

September–December 2022

Worked with Professor Jon Miller to process and analyze data from the Swift XRT; used NASA HEASARC software to characterize light curves, fluxes, and luminosities of several high-energy astrophysical sources

Selected Works

[8] “Chemical Divergence and Water Depletion: Gas Properties of Evolved Upper Scorpius Disks Revealed by JWST/MIRI”

Raul, E., et al., 2026, manuscript submitted to AJ

[7] “The JDISC Survey: Linking the Physics and Chemistry of Inner and Outer Protoplanetary Disk Zones”

Arulanantham, N., et al. (including **Raul, E.**), 2025, AJ, 170, 67

[6] “Escaping Helium, Depleted Methane, and Hints of Carbon Dioxide Suggest a High-Metallicity Atmosphere on Sub-Neptune GJ 3090 b from JWST Transit Spectroscopy”

Ahrer, E., et al. (including **Raul, E.**), 2025, ApJL, 985, L10

[5] “Tracking the Evolution of Hydrocarbons Through Carbon Grain Destruction in Protoplanetary Disks”

Raul, E., Alarcón, F., Bergin, E. A, 2025, ApJ, 982, 155

[4] “JWST/NIRISS Reveals the Water-rich “Steam World” Atmosphere of GJ 9827 d”

Piaulet-Ghorayeb, C., et al. (including **Raul, E.**), 2024, ApJL, 974, L10

[3] “JWST Reveals CH₄, CO₂, and H₂O in a Metal-rich Miscible Atmosphere on a Two-Earth-Radius Exoplanet”

Benneke, B., et al. (including **Raul, E.**), 2024, manuscript submitted to ApJ

[2] “Recent Swift Observations of NGC 5907 ULX-1”

Raul, E., Bacon, S., Brumback, M., Hoin, S., Miller, J. M., Reynolds, M. T., 2022, ATEL 15809

[1] “Swift Observations of the M51 ULX RX J133007+47110”

Raul, E., Brumback, M., Miller, J. M., Reynolds, M. T., et al., 2022, ATEL 15803

Selected Presentations

Accepted Talk, Discs on the Exe, Exeter, UK July 2026

Accepted Talk, Emerging Researchers in Exoplanet Science Symposium XI, Ann Arbor, MI July 2026

Contributed Talk, Cool Stars 23, Tokyo, Japan June 2026

Contributed Talk/**SOC**, Great Lakes Exoplanet Area Meeting, Madison, WI November 2025

Contributed Talk/Dwornik Judge, Lunar & Planetary Science Conference, The Woodlands, TX March 2025

Contributed Talk, Bay Area Exoplanet Meeting 47, San Francisco, CA December 2024

Contributed Poster, Astrobiology and the Future of Life Meeting, Houston, TX October 2024

Contributed Talk, Emerging Researchers in Exoplanet Science Symposium IX, Ithaca, NY July 2024

Contributed Talk, American Astronomical Society Meeting 243, New Orleans, LA January 2024

Contributed Lightning Talk, Great Lakes Exoplanet Area Meeting, Bloomington, IN October 2023

Honors and Awards

NASA Wisconsin Space Grant Consortium: Graduate & Professional Research Fellowship Summer 2026

Grant award: **\$4,381**

Blair Savage Astronomy Travel Fund Winter 2026

Grant award: **\$750**

University Honors/Dean’s List/James B. Angell Scholar Fall 2020-Winter 2024

Michigan Competitive Scholarship Fall 2020-Winter 2023

Frank J. Shay Scholarship Fall 2022-Winter 2023

Regents Merit Scholarship Fall 2020-Winter 2021

Public Talks and Outreach

Talk at Kettle Moraine State Park, WI	May 2026
Graduate Research Showcase at the Capitol, WI	April 2026
Science Night Presenter at Emerson Elementary School, Madison, WI	March 2026
Talk at Eisenhower Middle-High School, New Berlin, WI	January 2026
1st Grade Astronomy Unit Teacher at Van Hise Elementary School, Madison, WI	January 2026
Organizer of Stargazing at Science on the Square (Wisconsin Science Festival)	October 2025
Talk at Buckhorn State Park, WI	August 2025
Talk at Newport State Park, WI	August 2025
Talk at Monona Public Library, WI	August 2025
Science Volunteer at UW–Madison Wonders of Physics Fair	February 2025
Physics Demonstrations Volunteer at Metro Detroit Youth Clubs	November 2023

Academic Service

American Association for the Advancement of Science Mass Media Fellowship Location: MIT Technology Review	Summer 2026
American Astronomical Society (AAS) Congressional Visit Days Volunteer	April 2025
Skywatching Volunteer at Horwitz-DeRemer Planetarium	October 2025-
UW–Madison Department of Astronomy Telescope Lead	September 2025-
Washburn Observatory Volunteer	September 2024-

Science Communication and Press Releases

AAS - “Rocket Science Needs Rocket Fuel: An Early Scientist’s Experience on the Hill”	May 2025
Universe Today Podcast - “First Ever Steam World Found by JWST”	December 2024
University of Michigan - “JWST finds first exotic 'steam world' shrouded in water vapor”	October 2024

Accepted Observing Proposals

JWST Cycle 5, 57.6 hours, GO 12157 (PI: Benneke, B.)	2026
MIRI Clarity: Uncovering the Mysteries of Water Worlds and Sub-Neptunes in the Mid-Infrared	
JWST Cycle 4, 68.5 hours, GO 7135 (PI: Zhang, K.)	2025
Population Study of Chemistry in the First Million Years of Planet Formation: A MIRI Survey of Embedded Disks in Ophiuchus	

Science Teams

RODEO: Rossiter Observations Determining Exoplanet Obliquities (PI: Mann A.)	2025-
ACES: Architecture of Compact Exoplanetary Systems (PI: Becker J.)	2025-
JDISCS: JWST Disk Infrared Spectral Chemistry Survey (PIs: Salyk C., Banzatti A., Pontoppidan K.)	2024-
JWST Water Worlds Survey Team (Member, PI: Benneke B.)	2024-

Teaching

Graduate TA, ASTRON 170 (The Dark Side of the Universe)	January-May 2025
Undergraduate Grader, PHYSICS 406 (Statistical and Thermal Physics)	September-December 2023